

RESTAURANT POS & ORDER MANAGEMENT SYSTEM

Complete API Endpoints & Response Output Specification Guide

System Configuration & Base Parameters:

- **Base URL:** <http://localhost:8080>
- **Content-Type:** application/json
- **Authentication:** Bearer <JWT_TOKEN>
- **Total Modules:** 14 Modules (Module 0 to Module 13)
- **Total Endpoints Covered:** 108 Endpoints (Complete API Surface)

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MODULE 0: AUTHENTICATION & REGISTRATION (/api/auth)

1. System Admin Registration

Role: Standard Auth

POST

POST /api/auth/register

Headers: Content-Type: application/json

Request Body Payload:

```
{
  "username": "admin1",
  "email": "admin@restaurant.com",
  "password": "Password123!",
  "fullName": "Main System Admin",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

Response Output: Status 201 Created

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "username": "admin1",
  "email": "admin@restaurant.com",
  "fullName": "Main System Admin",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "isActive": true,
  "createdAt": "2026-08-09T10:00:00Z"
}
```

2. User Login (Obtain JWT Token)

Role: Standard Auth

POST

POST /api/auth/login

Headers: Content-Type: application/json

Request Body Payload:

```
{
  "email": "admin@restaurant.com",
  "password": "Password123!"
}
```

Response Output: Status 200 OK

```
{
  "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJhZG1pbjEiLCJyb2xlIjoiaURNSU4iLCJpYXQiOiE3MjMyNjQwMDB9...",
  "role": "ADMIN",
  "username": "admin1",
  "expiresIn": 86400
}
```

MODULE 1: ADMIN MANAGEMENT (/api/v1/admin) [Role: ADMIN]

1. Get All Admins

Role: Standard Auth

GET

GET /api/v1/admin

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "username": "admin1",
    "email": "admin@restaurant.com",
    "fullName": "Main System Admin",
    "isActive": true
  },
  {
    "id": "4fa85f64-5717-4562-b3fc-2c963f66afa7",
    "username": "outlet_admin_1",
    "email": "outlet1@restaurant.com",
    "fullName": "Outlet Manager Admin",
    "isActive": true
  }
]
```

2. Get Admin By ID

Role: Standard Auth

GET

GET /api/v1/admin/{id}

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Get Admin By ID' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

3. Create New Admin

Role: Standard Auth

POST

POST /api/v1/admin

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "username": "outlet_admin_1",
  "email": "outlet1@restaurant.com",
  "password": "Password123!",
  "fullName": "Outlet Manager Admin",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

Response Output: Status 201 Created

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "username": "admin1",
  "email": "admin@restaurant.com",
  "fullName": "Main System Admin",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "isActive": true,
  "createdAt": "2026-08-09T10:00:00Z"
}
```

4. Update Admin Details

Role: Standard Auth

PUT

PUT /api/v1/admin/{id}

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "username": "outlet_admin_1_updated",
  "email": "outlet1_updated@restaurant.com",
  "password": "NewPassword123!",
  "fullName": "Updated Admin Name",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Update Admin Details' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

5. Delete Admin

Role: Standard Auth

DELETE

DELETE /api/v1/admin/{id}

Headers: Authorization: Bearer

Response Output: **Status 204 No Content**

```
{
  "message": "Action completed successfully with no content returned"
}
```

6. Toggle Admin Active Status

Role: Standard Auth

PATCH

PATCH /api/v1/admin/{id}/toggle-active

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Toggle Admin Active Status' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

MODULE 2: STAFF & ACCESS CONTROL MANAGEMENT (/api/staff)

1. Get All Staff Members

Role: Standard Auth

GET

GET /api/staff

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "stf-101",
    "name": "Captain Rahul",
    "role": "CAPTAIN",
    "section": "Main Hall",
    "employmentType": "FULL_TIME",
    "isActive": true
  }
]
```

2. Get Staff Member By ID

Role: Standard Auth

GET

GET /api/staff/{id}

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Get Staff Member By ID' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

3. Register Staff Member

Role: ADMIN

POST

POST /api/staff

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "name": "Captain Rahul",
  "pin": "1234",
  "role": "CAPTAIN",
  "section": "Main Hall",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "employmentType": "FULL_TIME"
}
```

Response Output: Status 201 Created

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Register Staff Member' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

4. Staff PIN-based POS Login

Role: Standard Auth

POST

POST /api/staff/login-pin

Headers: Content-Type: application/json

Request Body Payload:

```
{
  "pin": "1234"
}
```

Response Output: Status 200 OK

```
{
  "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWUiOiJhZG1pbjEiLCJyb2xlIjoiQURNSU4iLCJpYXQiOiE3MjMyNjQwMDB9...",
  "role": "ADMIN",
  "username": "admin1",
  "expiresIn": 86400
}
```

5. Staff Shift Clock-In

Role: Standard Auth

POST

POST /api/staff/clock-in?name=Captain

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "logId": "att-998811",
  "staffName": "Captain Rahul",
  "clockInTime": "2026-08-09T08:00:00Z",
  "clockOutTime": null,
  "shiftDurationHours": null,
  "status": "ACTIVE"
}
```

6. Staff Shift Clock-Out

Role: Standard Auth

POST

POST /api/staff/clock-out?name=Captain

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "logId": "att-998811",
  "staffName": "Captain Rahul",
  "clockInTime": "2026-08-09T08:00:00Z",
  "clockOutTime": "2026-08-09T16:00:00Z",
  "shiftDurationHours": 8.0,
  "status": "COMPLETED"
}
```

7. Create Shift

Role: ADMIN

POST

POST /api/staff/shifts

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "shiftName": "Morning Shift",
  "startTime": "08:00:00",
  "endTime": "16:00:00"
}
```

Response Output: **Status 201 Created**

```
{
  "shiftId": "4fa85f64-5717-4562-b3fc-2c963f66afa7",
  "shiftName": "Morning Shift",
  "startTime": "08:00:00",
  "endTime": "16:00:00",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

8. Assign Staff to Shift

Role: MANAGER, ADMIN

POST

POST /api/staff/shifts/assign

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "staffId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "shiftId": "4fa85f64-5717-4562-b3fc-2c963f66afa7",
  "shiftDate": "2026-07-19"
}
```

Response Output: Status 201 Created

```
{
  "shiftId": "4fa85f64-5717-4562-b3fc-2c963f66afa7",
  "shiftName": "Morning Shift",
  "startTime": "08:00:00",
  "endTime": "16:00:00",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

9. Log Manager Audit Trail

Role: MANAGER

POST

POST /api/staff/audit?action=VOID_BILL&orderId;=&amount;=150.00&reason;=Customer

Headers: Authorization: Bearer

Response Output: Status 201 Created

```
{
  "customerId": "cust-881",
  "phone": "9876543210",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "loyaltyPoints": 50,
  "isVip": false
}
```

10. Get Staff Individual Performance Metrics

Role: MANAGER, ADMIN

GET

GET /api/staff/{id}/performance

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "staffId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "name": "Captain Rahul",
  "totalCoversServed": 142,
  "totalRevenueGenerated": 28400.5,
  "orderAccuracyPercentage": 98.5,
  "totalShiftHours": 160.0
}
```

11. Export Payroll Report (CSV Download)

Role: ADMIN

GET

GET /api/staff/payroll/export

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
Staff ID,Name,Role,Hours Worked,Hourly Rate,Total Salary
STF001,Captain Rahul,CAPTAIN,160,150.00,24000.00
STF002,Chef Priya,KITCHEN,180,200.00,36000.00
```

12. Update Staff Member

Role: ADMIN

PUT

PUT /api/staff/{id}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Update Staff Member' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

13. Delete Staff Member

Role: ADMIN

DELETE

DELETE /api/staff/{id}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Delete Staff Member' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

14. Toggle Staff Active Status

Role: ADMIN

PATCH

PATCH /api/staff/{id}/toggle-active

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Toggle Staff Active Status' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

MODULE 3: TABLE & WAITLIST MANAGEMENT (/api/v1/tables, /api/v1/waitlist)

1. Get Tables by Outlet

Role: CASHIER, CAPTAIN,
MANAGER

GET

GET /api/v1/tables/{out letId}

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "id": "tbl-001",
    "tableNumber": "T1",
    "capacity": 4,
    "section": "Main Hall",
    "status": "OCCUPIED",
    "xPos": 10,
    "yPos": 20,
    "covers": 3,
    "currentOrderId": "ord-7788"
  },
  {
    "id": "tbl-002",
    "tableNumber": "T2",
    "capacity": 2,
    "section": "Patio",
    "status": "AVAILABLE",
    "xPos": 30,
    "yPos": 20,
    "covers": 0
  }
]
```

2. Get All Tables

Role: CASHIER, CAPTAIN,
MANAGER, ADMIN

GET

GET /api/v1/tables

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "id": "tbl-001",
    "tableNumber": "T1",
    "capacity": 4,
    "section": "Main Hall",
    "status": "OCCUPIED",
    "xPos": 10,
    "yPos": 20,
    "covers": 3,
    "currentOrderId": "ord-7788"
  },
  {
    "id": "tbl-002",
    "tableNumber": "T2",
    "capacity": 2,
    "section": "Patio",
    "status": "AVAILABLE",
    "xPos": 30,
    "yPos": 20,
    "covers": 0
  }
]
```

3. Create Table

Role: ADMIN

POST

POST /api/v1/tables**Headers:** Authorization: Bearer**Request Body Payload:**

```
{
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "tableNumber": "T1",
  "capacity": 4,
  "section": "Main Hall",
  "status": "AVAILABLE",
  "xPos": 10,
  "yPos": 20
}
```

Response Output: Status 200 OK

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Create Table' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

4. Update Table Layout

Role: ADMIN

PUT

PUT /api/v1/tables/{id}**Headers:** Authorization: Bearer**Response Output: Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Update Table Layout' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

5. Delete Table

Role: ADMIN

DELETE

DELETE /api/v1/tables/{id}**Headers:** Authorization: Bearer**Response Output: Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Delete Table' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

6. Bulk Update Table Floor Layout

Role: ADMIN

PUT

PUT /api/v1/tables/bulk**Headers:** Authorization: Bearer**Response Output: Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Bulk Update Table Floor Layout' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

7. Assign Order to Table

Role: CAPTAIN, MANAGER

PUT

PUT /api/v1/tables/{id}/assign?orderId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Assign Order to Table' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

8. Update Table Status

Role: CASHIER, CAPTAIN, MANAGER

PUT

PUT /api/v1/tables/{id}/status?status=OCCUPIED

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Update Table Status' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

9. Merge Tables

Role: CAPTAIN, MANAGER

POST

POST /api/v1/tables/merge?primaryTableId=

Headers: Authorization: Bearer

Request Body Payload:

```
["<TABLE_2_ID>", "<TABLE_3_ID>"]
```

Response Output: **Status 200 OK**

```
{"status": "SUCCESS", "message": "Table action 'Merge Tables' completed successfully"}
```

10. Unmerge Tables

Role: CAPTAIN, MANAGER

POST

POST /api/v1/tables/unmerge?primaryTableId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{"status": "SUCCESS", "message": "Table action 'Unmerge Tables' completed successfully"}
```

11. Transfer Table Order

Role: CAPTAIN, MANAGER

POST

POST /api/v1/tables/transfer?sourceTableId=&destinationTableId;=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{"status": "SUCCESS", "message": "Table action 'Transfer Table Order' completed successfully"}
```

12. Update Table Covers Count

Role: CAPTAIN, MANAGER

PUT

PUT /api/v1/tables/{id}/covers?count=4

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Update Table Covers Count' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

13. Get Average Table Turn Time

Role: MANAGER, ADMIN

GET

GET /api/v1/tables/analytics/turn-time

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "averageTurnTimeMinutes": 42.5,
  "totalTablesTurnedToday": 38
}
```

14. Get Active Waitlist

Role: CASHIER, CAPTAIN,
MANAGER, ADMIN

GET

GET /api/v1/waitlist?outletId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "waitlistId": "wt-5544",
    "guestName": "John Doe",
    "guestPhone": "9876543210",
    "partySize": 4,
    "status": "WAITING",
    "createdAt": "2026-08-09T12:30:00Z"
  }
]
```

15. Add Walk-in Guest to Waitlist

Role: CASHIER, CAPTAIN,
MANAGER

POST

POST /api/v1/waitlist

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "guestName": "John Doe",
  "guestPhone": "9876543210",
  "partySize": 4
}
```

Response Output: **Status 201 Created**

```
{
  "waitlistId": "wt-5544",
  "guestName": "John Doe",
  "guestPhone": "9876543210",
  "partySize": 4,
  "estimatedWaitTimeMinutes": 15,
  "status": "WAITING"
}
```

16. Seat Guest from Waitlist

Role: CAPTAIN, MANAGER

PUT

PUT /api/v1/waitlist/{id}/seat?tableId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "waitlistId": "wt-5544",
    "guestName": "John Doe",
    "guestPhone": "9876543210",
    "partySize": 4,
    "status": "WAITING",
    "createdAt": "2026-08-09T12:30:00Z"
  }
]
```

17. Cancel Waitlist Entry

Role: CASHIER, CAPTAIN,
MANAGER

PUT

PUT /api/v1/waitlist/{id}/cancel

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "waitlistId": "wt-5544",
    "guestName": "John Doe",
    "guestPhone": "9876543210",
    "partySize": 4,
    "status": "WAITING",
    "createdAt": "2026-08-09T12:30:00Z"
  }
]
```

MODULE 4: MENU, CATEGORIES, MODIFIERS, COMBOS & STATIONS (/api/v1/menu)

1. Get All Categories

Role: Standard Auth

GET

GET /api/v1/menu/categories

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "cat-01",
    "name": "Starters",
    "description": "Appetizers and snacks",
    "displayOrder": 1
  },
  {
    "id": "cat-02",
    "name": "Main Course",
    "description": "Curries and Indian breads",
    "displayOrder": 2
  }
]
```

2. Create Category

Role: ADMIN, MANAGER

POST

POST /api/v1/menu/categories

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "name": "Starters",
  "description": "Appetizers and snacks",
  "displayOrder": 1
}
```

Response Output: **Status 200 OK**

```
[
  {
    "id": "cat-01",
    "name": "Starters",
    "description": "Appetizers and snacks",
    "displayOrder": 1
  },
  {
    "id": "cat-02",
    "name": "Main Course",
    "description": "Curries and Indian breads",
    "displayOrder": 2
  }
]
```

3. Get All Menu Items

Role: Standard Auth

GET

GET /api/v1/menu/items

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

4. Get Menu Items by Category

Role: Standard Auth

GET

GET /api/v1/menu/items/category/{categoryId}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

5. Get Active Time-Based Menu Items

Role: Standard Auth

GET

GET /api/v1/menu/items/active?time=13:30:00

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

6. Create Menu Item

Role: ADMIN, MANAGER

POST

POST /api/v1/menu/items**Headers:** Authorization: Bearer**Request Body Payload:**

```
{
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "name": "Paneer Butter Masala",
  "price": 280.00,
  "gstRate": 5.0,
  "foodType": "Veg",
  "isAvailable": true,
  "station": "HOT"
}
```

Response Output: Status 200 OK

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

7. Update Menu Item

Role: ADMIN, MANAGER

PUT

PUT /api/v1/menu/items/{id}**Headers:** Authorization: Bearer**Response Output: Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

8. Toggle Item Availability

Role: ADMIN, MANAGER

PATCH

PATCH /api/v1/menu/items/{id}/toggle-availability

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

9. Mark Item 86'd / Available

Role: ADMIN, MANAGER, KITCHEN

PATCH

PATCH /api/v1/menu/items/{id}/86?available=false

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

10. Get Daily Specials

Role: Standard Auth

GET

GET /api/v1/menu/items/specials

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

11. Set Daily Special

Role: ADMIN, MANAGER

PUT

PUT /api/v1/menu/items/{id}/special?special=true&specialPrice;=250.00

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

12. Get Modifier Groups for Item

Role: Standard Auth

GET

GET /api/v1/menu/items/{itemId}/modifier-groups

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

13. Create Modifier Group

Role: ADMIN, MANAGER

POST

POST /api/v1/menu/modifier-groups

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "menuItemId": "<ITEM_ID>",
  "name": "Spice Level",
  "minSelection": 1,
  "maxSelection": 1
}
```

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Create Modifier Group' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

14. Get Options for Modifier Group

Role: Standard Auth

GET

GET /api/v1/menu/modifier-groups/{groupId}/options

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Get Options for Modifier Group' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

15. Create Modifier Option

Role: ADMIN, MANAGER

POST

POST /api/v1/menu/modifier-options

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "groupId": "<GROUP_ID>",
  "name": "Extra Spicy",
  "price": 20.00
}
```

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Create Modifier Option' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

16. Get Combo Components

Role: Standard Auth

GET

GET /api/v1/menu/items/{comboItemId}/combo-components

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "itm-201",
    "name": "Paneer Butter Masala",
    "price": 280.0,
    "gstRate": 5.0,
    "foodType": "Veg",
    "isAvailable": true,
    "is86ed": false,
    "station": "HOT"
  }
]
```

17. Add Combo Component

Role: ADMIN, MANAGER

POST

POST /api/v1/menu/combo-components

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "message": "Endpoint 'Add Combo Component' executed successfully",
  "timestamp": "2026-08-09T16:35:00Z"
}
```

18. Get Digital QR Code (PNG Image)

Role: Standard Auth

GET

GET /api/v1/menu/outlets/{outletId}/qr**Headers:** Authorization: Bearer**Response Output:** Status 200 OK

[Binary PNG Image Data: Base64-encoded Digital QR Code PNG Stream]

19. Get Kitchen Stations

Role: Standard Auth

GET

GET /api/v1/kitchen-stations**Headers:** Authorization: Bearer**Response Output:** Status 200 OK

```
[
  {
    "id": "stn-1",
    "name": "GRILL"
  },
  {
    "id": "stn-2",
    "name": "HOT_KITCHEN"
  },
  {
    "id": "stn-3",
    "name": "BAR"
  }
]
```

20. Create Kitchen Station

Role: ADMIN

POST

POST /api/v1/kitchen-stations**Headers:** Authorization: Bearer**Request Body Payload:**

```
{ "name": "GRILL" }
```

Response Output: Status 200 OK

```
[
  {
    "id": "stn-1",
    "name": "GRILL"
  },
  {
    "id": "stn-2",
    "name": "HOT_KITCHEN"
  },
  {
    "id": "stn-3",
    "name": "BAR"
  }
]
```

21. Delete Kitchen Station

Role: ADMIN

DELETE

DELETE /api/v1/kitchen-stations/{id}

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "id": "stn-1",
    "name": "GRILL"
  },
  {
    "id": "stn-2",
    "name": "HOT_KITCHEN"
  },
  {
    "id": "stn-3",
    "name": "BAR"
  }
]
```

MODULE 5: ORDER PROCESSING & KOT MANAGEMENT (/api/v1/orders)

1. Get All Orders

Role: Standard Auth

GET

GET /api/v1/orders

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

2. Get Order By ID

Role: Standard Auth

GET

GET /api/v1/orders/{id}

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

3. Get Order Items

Role: Standard Auth

GET

GET /api/v1/orders/{id}/items

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

4. Create New Order

Role: CASHIER, CAPTAIN,
MANAGER

POST

POST /api/v1/orders

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "order": {
    "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "tableId": "<TABLE_ID>",
    "orderType": "DINE_IN",
    "covers": 2
  },
  "items": [
    {
      "menuItem": "<ITEM_ID>",
      "quantity": 2,
      "unitPrice": 150.00,
      "course": "Main Course"
    }
  ]
}
```

Response Output: **Status 201 Created**

```
{
  "orderId": "ord-889900",
  "orderNumber": "#1042",
  "tableId": "tbl-001",
  "orderType": "DINE_IN",
  "status": "PLACED",
  "totalAmount": 300.0,
  "createdAt": "2026-08-09T13:00:00Z"
}
```

5. Fire KOT to Kitchen

Role: CASHIER, CAPTAIN,
MANAGER

PUT

PUT /api/v1/orders/{id}/fire-kot

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{"status": "SUCCESS", "message": "Action 'Fire KOT to Kitchen' processed successfully"}
```

6. Fire Fine Dining Course Sequence

Role: CASHIER, CAPTAIN,
MANAGER

PUT

PUT /api/v1/orders/{id}/fire-course?course=Starters

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{"status": "SUCCESS", "message": "Action 'Fire Fine Dining Course Sequence' processed successfully"}
```

7. Void Order Item (Requires Manager PIN)

Role: CASHIER, CAPTAIN,
MANAGER

DELETE

DELETE /api/v1/orders/items/{itemId}?managerPin=9999

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

8. Get Draft Orders by Table

Role: Standard Auth

GET

GET /api/v1/orders/drafts/table/{tableId}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

9. Get Draft Orders by Token

Role: Standard Auth

GET

GET /api/v1/orders/drafts/token/{token}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

10. Modify Order Items

Role: CAPTAIN, MANAGER

PUT

PUT /api/v1/orders/{id}/modify?managerPin=9999**Headers:** Authorization: Bearer**Request Body Payload:**

List of OrderItem objects

Response Output: Status 200 OK

```
{"status": "SUCCESS", "message": "Action 'Modify Order Items' processed successfully"}
```

11. Split Order Items

Role: CASHIER, CAPTAIN,
MANAGER

POST

POST /api/v1/orders/{id}/split**Headers:** Authorization: Bearer**Request Body Payload:**

List of OrderItemSplitRequest objects

Response Output: Status 200 OK

```
{"status": "SUCCESS", "message": "Action 'Split Order Items' processed successfully"}
```

12. Apply Item-Level Discount

Role: CAPTAIN, MANAGER

PUT

PUT /api/v1/orders/items/{orderId}/discount?discountAmount=20.00&managerPin;=9999**Headers:** Authorization: Bearer**Response Output: Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

13. Mark Item Complimentary

Role: CAPTAIN, MANAGER

PUT

PUT /api/v1/orders/items/{orderId}/complimentary?managerPin=9999**Headers:** Authorization: Bearer**Response Output: Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

14. Get Order Lifecycle Audit Timeline

Role: CAPTAIN, MANAGER, ADMIN

GET

GET /api/v1/orders/{id}/timeline

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "timestamp": "2026-08-09T13:00:00Z",
    "event": "ORDER_CREATED",
    "performedBy": "Captain Rahul"
  },
  {
    "timestamp": "2026-08-09T13:02:00Z",
    "event": "KOT_FIRED_HOT_KITCHEN",
    "performedBy": "Captain Rahul"
  },
  {
    "timestamp": "2026-08-09T13:12:00Z",
    "event": "ITEM_BUMPED_READY",
    "performedBy": "Chef Priya"
  }
]
```

MODULE 6: KITCHEN DISPLAY SYSTEM - KDS (/api/v1/kds) [Role: KITCHEN]

1. Get Active KOTs by Kitchen Station

Role: Standard Auth

GET

GET /api/v1/kds/{stationId}/orders

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

2. Bump KOT Item (Mark Prepared)

Role: Standard Auth

PUT

PUT /api/v1/kds/items/{itemId}/bump

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{"status": "SUCCESS", "message": "Action 'Bump KOT Item (Mark Prepared)' processed successfully"}
```

3. Get Last 5 Recalled Bumped Orders

Role: Standard Auth

GET

GET /api/v1/kds/{stationId}/recalled

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

4. Get Kitchen Preparation Performance Metrics

Role: Standard Auth

GET

GET /api/v1/kds/performance?stationId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

5. Sync Offline Bump Actions

Role: Standard Auth

POST

POST /api/v1/kds/sync

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "bumpedItemIds": ["<ITEM_1_ID>", "<ITEM_2_ID>"]
}
```

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

MODULE 7: PAYMENT PROCESSING, BILLING & RECEIPTS (/api/v1)

1. Get All Bills

Role: Standard Auth

GET

GET /api/v1/bills

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "billId": "bil-990011",
    "invoiceNumber": "INV-2026-0891",
    "netAmount": 315.0,
    "paymentStatus": "SETTLED"
  }
]
```

2. Get Bill By ID

Role: Standard Auth

GET

GET /api/v1/bills/{id}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "billId": "bil-990011",
    "invoiceNumber": "INV-2026-0891",
    "netAmount": 315.0,
    "paymentStatus": "SETTLED"
  }
]
```

3. Generate GST Invoice

Role: CASHIER

POST

POST /api/v1/bills?orderId=&discountAmount;=0&customerPhone;=9876543210&pointsToRedeem;=50&managerPin;=9999

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "UNPAID"
}
```

4. Settle Bill (Single Payment Method)

Role: CASHIER

POST

POST /api/v1/bills/{id}/settle?customerPhone=9876543210**Headers:** Authorization: Bearer**Response Output:** Status 201 Created

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "SETTLED"
}
```

5. Settle Bill Split (Multiple Payment Methods up to 4)

Role: CASHIER

POST

POST /api/v1/bills/{id}/settle-split**Headers:** Authorization: Bearer**Request Body Payload:**

```
{
  "payments": [
    { "method": "CASH", "amount": 200.00 },
    { "method": "UPI", "amount": 115.00 }
  ],
  "customerPhone": "9876543210",
  "tipAmount": 20.00
}
```

Response Output: Status 201 Created

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "SETTLED"
}
```

6. Split Bill Equally Across Guests

Role: CASHIER

POST

POST /api/v1/bills/split/equal?orderId=&numGuests;=3&managerPin;=9999**Headers:** Authorization: Bearer**Response Output:** Status 201 Created

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "UNPAID"
}
```

7. Split Bill by Item Selection

Role: CASHIER

POST

POST /api/v1/bills/split/items?orderId=

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "itemGroups": [
    ["<ITEM_1_ID>", "<ITEM_2_ID>"],
    ["<ITEM_3_ID>"]
  ],
  "managerPin": "9999"
}
```

Response Output: Status 201 Created

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "UNPAID"
}
```

8. Get Formatted Thermal Receipt Text

Role: CASHIER, MANAGER

GET

GET /api/v1/bills/{id}/receipt

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
=====
                ROYAL FINE DINING POS
                GSTIN: 29AAAAA0000A1Z5
=====
Receipt #: REC-4091      Date: 09/08/2026
Table: T1                Server: Rahul
-----
Paneer Butter Masala x 2    560.00
Butter Naan x 4             160.00
-----
Subtotal:                  720.00
CGST (2.5%):               18.00
SGST (2.5%):               18.00
Total Payable:             756.00
=====
                Thank you for dining with us!
```

9. Send Digital Receipt via SMS/WhatsApp

Role: CASHIER, MANAGER

POST

POST /api/v1/bills/{id}/send-digital?phoneNumber=9876543210

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "UNPAID"
}
```

10. Initiate Razorpay/Pine Labs Payment

Role: CASHIER

POST

POST /api/v1/payments/razorpay?orderId=&amount;=315.00

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "UNPAID"
}
```

11. Void Bill and Process Refund

Role: MANAGER

POST

POST /api/v1/bills/{id}/void?managerPin=9999

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "billId": "bil-990011",
  "invoiceNumber": "INV-2026-0891",
  "orderId": "ord-889900",
  "subTotal": 300.0,
  "taxTotal": 15.0,
  "discountTotal": 0.0,
  "netAmount": 315.0,
  "paymentStatus": "UNPAID"
}
```

12. Get Day-End Z-Report

Role: MANAGER

GET

GET /api/v1/reports/z-report?date=2026-07-18&managerId;=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "reportDate": "2026-07-18",
  "grossSales": 45200.0,
  "netSales": 43047.62,
  "totalTax": 2152.38,
  "cashPayments": 12000.0,
  "upiPayments": 23500.0,
  "cardPayments": 9700.0,
  "voidedBillsTotal": 350.0
}
```

13. Reconcile Cash Drawer

Role: MANAGER

GET

GET /api/v1/reports/cash-drawer-reconciliation?date=2026-07-18&startingCash;=2000.00

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "reportDate": "2026-07-18",
  "grossSales": 45200.0,
  "netSales": 43047.62,
  "totalTax": 2152.38,
  "cashPayments": 12000.0,
  "upiPayments": 23500.0,
  "cardPayments": 9700.0,
  "voidedBillsTotal": 350.0
}
```

MODULE 8: ONLINE ORDER AGGREGATION - SWIGGY / ZOMATO (/api/v1/aggregator)

1. Receive Aggregator Order Webhook

Role: CASHIER, CAPTAIN,
MANAGER

POST

POST /api/v1/aggregator/webhook

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "aggregator": "SWIGGY",
  "aggregatorOrderId": "SW-99881122",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "items": [
    { "menuItemName": "Paneer Butter Masala", "quantity": 1, "price": 280.00 }
  ]
}
```

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "aggregatorOrderId": "SW-99881122",
  "systemOrderId": "ord-556677",
  "estimatedPrepTimeMinutes": 25
}
```

2. Confirm Order Prep Time to Aggregator

Role: CASHIER, CAPTAIN,
MANAGER

PUT

PUT /api/v1/aggregator/orders/{id}/accept?prepTime=25

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

3. Modify Aggregator Order Item / Out of Stock

Role: MANAGER

PUT

PUT /api/v1/aggregator/orders/{orderId}/cancel-item/{itemId}?managerPin=9999

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "orderId": "ord-889900",
    "orderNumber": "#1042",
    "tableId": "tbl-001",
    "status": "IN_KITCHEN",
    "itemsCount": 2,
    "totalAmount": 300.0
  }
]
```

4. Receive Delivery Rider Arrival ETA Update

Role: CASHIER, MANAGER

POST

POST /api/v1/aggregator/rider-update

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "orderId": "<ORDER_ID>",
  "riderName": "Ramesh Kumar",
  "riderPhone": "9811223344",
  "etaMinutes": 10
}
```

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "aggregatorOrderId": "SW-99881122",
  "systemOrderId": "ord-556677",
  "estimatedPrepTimeMinutes": 25
}
```

5. Reconcile Aggregator Payout vs Settlements

Role: MANAGER

POST

POST /api/v1/aggregator/reconcile?aggregator=SWIGGY&date;=2026-07-18&actualPayout;=4500.00

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "status": "SUCCESS",
  "aggregatorOrderId": "SW-99881122",
  "systemOrderId": "ord-556677",
  "estimatedPrepTimeMinutes": 25
}
```

6. Get Aggregator Performance Analytics

Role: MANAGER, ADMIN

GET

GET /api/v1/aggregator/analytics

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "totalAggregatorOrders": 124,
  "swiggyRevenue": 34500.0,
  "zomatoRevenue": 28900.0,
  "cancellationRate": 1.2
}
```

MODULE 9: INVENTORY & SUPPLIER MANAGEMENT (/api/v1/inventory) [Role: MANAGER]

1. Get Current Stock Master

Role: Standard Auth

GET

GET /api/v1/inventory

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "ingredientId": "ing-01",
    "name": "Fresh Cottage Cheese (Paneer)",
    "unit": "KG",
    "availableQty": 12.5,
    "costPricePerUnit": 320.0
  }
]
```

2. Record Goods Received Note (GRN) Stock Increase

Role: Standard Auth

POST

POST /api/v1/inventory/grn?ingredientId=&quantityReceived;=10.5&costPrice;=120.00

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "inv-rec-991",
  "message": "Inventory record updated successfully"
}
```

3. Get Low-Stock Alerts

Role: Standard Auth

GET

GET /api/v1/inventory/alerts

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "ingredientName": "Paneer",
    "currentStock": 2.5,
    "unit": "KG",
    "minReorderLevel": 5.0,
    "status": "LOW_STOCK"
  },
  {
    "ingredientName": "Amul Butter",
    "currentStock": 1.0,
    "unit": "KG",
    "minReorderLevel": 3.0,
    "status": "LOW_STOCK"
  }
]
```

4. Create Supplier Profile

Role: Standard Auth

POST

POST /api/v1/inventory/suppliers

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "supplierName": "Fresh Dairy Farms",
  "contactNumber": "9876500000",
  "itemsSupplied": "Milk, Butter, Paneer"
}
```

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "inv-rec-991",
  "message": "Inventory record updated successfully"
}
```

5. Get All Suppliers

Role: Standard Auth

GET

GET /api/v1/inventory/suppliers

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "ingredientId": "ing-01",
    "name": "Fresh Cottage Cheese (Paneer)",
    "unit": "KG",
    "availableQty": 12.5,
    "costPricePerUnit": 320.0
  }
]
```

6. Confirm Daily Opening Stock

Role: Standard Auth

POST

POST /api/v1/inventory/daily-stock/opening**Headers:** Authorization: Bearer**Request Body Payload:**

List of DailyStockSetupRequest objects

Response Output: Status 201 Created

```
{
  "status": "SUCCESS",
  "id": "inv-rec-991",
  "message": "Inventory record updated successfully"
}
```

7. Register Daily Closing Stock (Waste Variance Tracking)

Role: Standard Auth

POST

POST /api/v1/inventory/daily-stock/closing**Headers:** Authorization: Bearer**Request Body Payload:**

List of DailyStockCloseRequest objects

Response Output: Status 201 Created

```
{
  "status": "SUCCESS",
  "id": "inv-rec-991",
  "message": "Inventory record updated successfully"
}
```

8. Raise Purchase Order (PO) to Supplier

Role: Standard Auth

POST

POST /api/v1/inventory/purchase-orders**Headers:** Authorization: Bearer**Request Body Payload:**

```
{
  "supplierId": "<SUPPLIER_ID>",
  "ingredientId": "<INGREDIENT_ID>",
  "orderedQuantity": 50.0
}
```

Response Output: Status 201 Created

```
{
  "orderId": "ord-889900",
  "orderNumber": "#1042",
  "tableId": "tbl-001",
  "orderType": "DINE_IN",
  "status": "PLACED",
  "totalAmount": 300.0,
  "createdAt": "2026-08-09T13:00:00Z"
}
```

9. Receive Purchase Order GRN

Role: Standard Auth

POST

POST /api/v1/inventory/purchase-orders/{id}/receive

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "orderId": "ord-889900",
  "orderNumber": "#1042",
  "tableId": "tbl-001",
  "orderType": "DINE_IN",
  "status": "PLACED",
  "totalAmount": 300.0,
  "createdAt": "2026-08-09T13:00:00Z"
}
```

10. Get Cost of Goods Sold (COGS) & Gross Margin Report

Role: Standard Auth

GET

GET /api/v1/inventory/reports/cogs

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "billId": "bil-990011",
    "invoiceNumber": "INV-2026-0891",
    "netAmount": 315.0,
    "paymentStatus": "SETTLED"
  }
]
```

11. Get Perishable Inventory Batches (FIFO Expiry Tracking)

Role: Standard Auth

GET

GET /api/v1/inventory/batches

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "ingredientId": "ing-01",
    "name": "Fresh Cottage Cheese (Paneer)",
    "unit": "KG",
    "availableQty": 12.5,
    "costPricePerUnit": 320.0
  }
]
```

MODULE 10: CUSTOMER CRM, LOYALTY & SMS CAMPAIGNS (/api/customers)

1. Get All Customer Profiles

Role: Standard Auth

GET

GET /api/customers

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "customerId": "cust-881",
    "phone": "9876543210",
    "name": "Jane Doe",
    "loyaltyPoints": 50
  }
]
```

2. Get Customer Profile By ID

Role: Standard Auth

GET

GET /api/customers/{id}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "customerId": "cust-881",
    "phone": "9876543210",
    "name": "Jane Doe",
    "loyaltyPoints": 50
  }
]
```

3. Create Customer Profile

Role: Standard Auth

POST

POST /api/customers

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "phone": "9876543210",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "dateOfBirth": "1995-08-25"
}
```

Response Output: **Status 201 Created**

```
{
  "customerId": "cust-881",
  "phone": "9876543210",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "loyaltyPoints": 50,
  "isVip": false
}
```

4. Register Customer with OTP Verification

Role: Standard Auth

POST

POST /api/customers/register-otp

Headers: Authorization: Bearer

Request Body Payload:

Same as CustomerRequest

Response Output: Status 201 Created

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "username": "admin1",
  "email": "admin@restaurant.com",
  "fullName": "Main System Admin",
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "isActive": true,
  "createdAt": "2026-08-09T10:00:00Z"
}
```

5. Verify Customer Registration OTP

Role: Standard Auth

POST

POST /api/customers/verify-otp

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "phone": "9876543210",
  "otpCode": "123456"
}
```

Response Output: Status 201 Created

```
{
  "customerId": "cust-881",
  "phone": "9876543210",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "loyaltyPoints": 50,
  "isVip": false
}
```

6. Update Customer Profile

Role: Standard Auth

PUT

PUT /api/customers/{id}

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "customerId": "cust-881",
    "phone": "9876543210",
    "name": "Jane Doe",
    "loyaltyPoints": 50
  }
]
```

7. Delete Customer Profile

Role: Standard Auth

DELETE

DELETE /api/customers/{id}

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "customerId": "cust-881",
    "phone": "9876543210",
    "name": "Jane Doe",
    "loyaltyPoints": 50
  }
]
```

8. Submit Customer Post-Visit Feedback Rating

Role: Standard Auth

POST

POST /api/customers/feedback

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "phone": "9876543210",
  "rating": 5,
  "comments": "Excellent food and service!"
}
```

Response Output: **Status 201 Created**

```
{
  "customerId": "cust-881",
  "phone": "9876543210",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "loyaltyPoints": 50,
  "isVip": false
}
```

9. Toggle VIP Customer Tag

Role: MANAGER

PUT

PUT /api/customers/{id}/vip?isVip=true

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "customerId": "cust-881",
    "phone": "9876543210",
    "name": "Jane Doe",
    "loyaltyPoints": 50
  }
]
```

10. Get Customer Visit & Spend Analytics

Role: MANAGER, ADMIN

GET

GET /api/customers/{id}/analytics

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "customerId": "cust-881",
  "totalVisits": 14,
  "totalLifetimeSpend": 12400.0,
  "averageOrderValue": 885.7,
  "loyaltyPointsBalance": 240,
  "isVip": true
}
```

11. Dispatch Targeted Promotional SMS Campaign

Role: MANAGER

POST

POST /api/customers/campaign?segment=HIGH_VALUE&messageText;=Special

Headers: Authorization: Bearer

Response Output: Status 201 Created

```
{
  "customerId": "cust-881",
  "phone": "9876543210",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "loyaltyPoints": 50,
  "isVip": false
}
```

12. Trigger Birthday Complimentary Offer SMS Dispatch

Role: MANAGER

POST

POST /api/customers/birthday-campaign

Headers: Authorization: Bearer

Response Output: Status 201 Created

```
{
  "customerId": "cust-881",
  "phone": "9876543210",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "loyaltyPoints": 50,
  "isVip": false
}
```

MODULE 11: RESERVATION MANAGEMENT (/api/v1/reservations)

1. Get All Reservations

Role: Standard Auth

GET

GET /api/v1/reservations

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "reservationId": "res-7722",
    "guestName": "Robert Smith",
    "partySize": 4,
    "status": "CONFIRMED"
  }
]
```

2. Create Phone/Walk-in Reservation

Role: MANAGER, ADMIN

POST

POST /api/v1/reservations**Headers:** Authorization: Bearer**Request Body Payload:**

```
{
  "outletId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "guestName": "Robert Smith",
  "guestPhone": "9812345678",
  "partySize": 4,
  "date": "2026-07-20",
  "time": "19:30:00",
  "specialRequests": "Window table requested",
  "dietaryRequirements": "Nut allergy",
  "preOrderItems": "Paneer Tikka"
}
```

Response Output: Status 201 Created

```
{
  "reservationId": "res-7722",
  "guestName": "Robert Smith",
  "guestPhone": "9812345678",
  "partySize": 4,
  "date": "2026-07-20",
  "time": "19:30:00",
  "status": "CONFIRMED"
}
```

3. Online Self-Service Table Reservation

Role: Standard Auth

POST

POST /api/v1/reservations/online**Headers:** Authorization: Bearer**Request Body Payload:**

Same as Reservation object

Response Output: Status 201 Created

```
{
  "reservationId": "res-7722",
  "guestName": "Robert Smith",
  "guestPhone": "9812345678",
  "partySize": 4,
  "date": "2026-07-20",
  "time": "19:30:00",
  "status": "CONFIRMED"
}
```

4. Guest Self-Service Cancellation via Link

Role: Standard Auth

POST

POST /api/v1/reservations/{id}/cancel-by-guest**Headers:** Authorization: Bearer**Response Output: Status 201 Created**

```
{
  "reservationId": "res-7722",
  "guestName": "Robert Smith",
  "guestPhone": "9812345678",
  "partySize": 4,
  "date": "2026-07-20",
  "time": "19:30:00",
  "status": "CONFIRMED"
}
```

5. Restaurant Cancellation

Role: MANAGER, ADMIN

POST

POST /api/v1/reservations/{id}/cancel

Headers: Authorization: Bearer

Response Output: Status 201 Created

```
{
  "reservationId": "res-7722",
  "guestName": "Robert Smith",
  "guestPhone": "9812345678",
  "partySize": 4,
  "date": "2026-07-20",
  "time": "19:30:00",
  "status": "CONFIRMED"
}
```

6. Update Reservation Status

Role: MANAGER, ADMIN

PUT

PUT /api/v1/reservations/{id}/status?status=SEATED

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "reservationId": "res-7722",
    "guestName": "Robert Smith",
    "partySize": 4,
    "status": "CONFIRMED"
  }
]
```

7. Get Reservation Analytics & No-Show Tracking

Role: MANAGER, ADMIN

GET

GET /api/v1/reservations/analytics

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "totalReservations": 86,
  "fulfilledSeatedPercentage": 91.5,
  "noShowRatePercentage": 4.2,
  "averagePartySize": 3.8
}
```

MODULE 12: MULTI-OUTLET CHAIN MANAGEMENT (/api/v1/chain)

1. Create Brand

Role: ADMIN

POST

POST /api/v1/chain/brands

Headers: Authorization: Bearer

Request Body Payload:

```
{ "name": "Royal Fine Dining Chain" }
```

Response Output: Status 201 Created

```
{
  "status": "SUCCESS",
  "id": "chn-obj-88",
  "message": "Chain operation executed successfully"
}
```

2. Create Outlet Branch

Role: ADMIN

POST

POST /api/v1/chain/outlets

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "brandId": "<BRAND_ID>",
  "name": "Downtown Outlet",
  "address": "MG Road, Suite 100",
  "gstIn": "29AAAAA000A1Z5"
}
```

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "chn-obj-88",
  "message": "Chain operation executed successfully"
}
```

3. Get All Chain Outlets

Role: Standard Auth

GET

GET /api/v1/chain/outlets

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "outletId": "out-101",
    "name": "Downtown Branch",
    "address": "MG Road, Suite 100",
    "isActive": true
  }
]
```

4. Push Central Master Menu to Outlets

Role: ADMIN

POST

POST /api/v1/chain/menu/push

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "brandId": "<BRAND_ID>",
  "outletIds": ["<OUTLET_1_ID>", "<OUTLET_2_ID>"]
}
```

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "chn-obj-88",
  "message": "Chain operation executed successfully"
}
```

5. Save Outlet-Specific Price/Availability Overrides

Role: MANAGER, ADMIN

POST

POST /api/v1/chain/menu/override

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "outletId": "<OUTLET_ID>",
  "menuItemId": "<ITEM_ID>",
  "overridePrice": 320.00,
  "isAvailable": true
}
```

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "chn-obj-88",
  "message": "Chain operation executed successfully"
}
```

6. Broadcast 86'd Item Unavailability Chain-wide

Role: ADMIN

POST

POST /api/v1/chain/menu/broadcast-unavailability?menuItemName=Salmon&isAvailable;=false

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "chn-obj-88",
  "message": "Chain operation executed successfully"
}
```

7. Create Brand-wide Promotional Offer

Role: ADMIN

POST

POST /api/v1/chain/promotions

Headers: Authorization: Bearer

Request Body Payload:

```
{
  "title": "Monsoon Festival 15% OFF",
  "discountPercentage": 15.0
}
```

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "chn-obj-88",
  "message": "Chain operation executed successfully"
}
```

8. Enable/Disable Mystery Audit Mode

Role: ADMIN

PUT

PUT /api/v1/chain/outlets/{id}/mystery-audit?active=true

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "outletId": "out-101",
    "name": "Downtown Branch",
    "address": "MG Road, Suite 100",
    "isActive": true
  }
]
```

9. Broadcast Corporate Alerts to Outlet Staff

Role: ADMIN

POST

POST /api/v1/chain/notifications/broadcast?brandId=&title;=System

Headers: Authorization: Bearer

Response Output: **Status 201 Created**

```
{
  "status": "SUCCESS",
  "id": "chn-obj-88",
  "message": "Chain operation executed successfully"
}
```

10. Calculate Franchise Royalty (GMV % Basis)

Role: ADMIN

GET

GET /api/v1/chain/outlets/{id}/royalty

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "brandName": "Royal Fine Dining Chain",
  "totalOutlets": 4,
  "totalSystemGMV": 4850000.0,
  "royaltyPercentage": 5.0,
  "calculatedRoyaltyFees": 242500.0
}
```

11. Get Chain Consolidated Financial Report

Role: ADMIN

GET

GET /api/v1/chain/reports/consolidated

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "billId": "bil-990011",
    "invoiceNumber": "INV-2026-0891",
    "netAmount": 315.0,
    "paymentStatus": "SETTLED"
  }
]
```

12. Get Outlet Side-by-Side Comparison Benchmarks

Role: ADMIN

GET

GET /api/v1/chain/reports/comparison

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
[
  {
    "billId": "bil-990011",
    "invoiceNumber": "INV-2026-0891",
    "netAmount": 315.0,
    "paymentStatus": "SETTLED"
  }
]
```

MODULE 13: ANALYTICS & REPORTING DASHBOARDS (/api/v1/analytics)

1. Get Daily Sales Summary Dashboard

Role: MANAGER, ADMIN

GET

GET /api/v1/analytics/daily-sales?startDate=2026-07-01&endDate;=2026-07-18&outletId;=

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "period": "2026-07-01 to 2026-07-18",
  "totalRevenue": 284500.0,
  "totalOrdersProcessed": 1420,
  "topSellingCategory": "Starters",
  "averageCheckAmount": 200.35,
  "dineInVsTakeawayRatio": "70:30"
}
```

2. Get Mid-Shift X-Report Live Snapshot

Role: CASHIER, MANAGER, ADMIN

GET

GET /api/v1/analytics/x-report?outletId=

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "period": "2026-07-01 to 2026-07-18",
  "totalRevenue": 284500.0,
  "totalOrdersProcessed": 1420,
  "topSellingCategory": "Starters",
  "averageCheckAmount": 200.35,
  "dineInVsTakeawayRatio": "70:30"
}
```

3. Generate End-of-Day Z-Report Settlement

Role: MANAGER, ADMIN

POST

POST /api/v1/analytics/z-report?date=2026-07-18&outletId;=&managerId;=

Headers: Authorization: Bearer

Response Output: Status 200 OK

```
{
  "period": "2026-07-01 to 2026-07-18",
  "totalRevenue": 284500.0,
  "totalOrdersProcessed": 1420,
  "topSellingCategory": "Starters",
  "averageCheckAmount": 200.35,
  "dineInVsTakeawayRatio": "70:30"
}
```

4. Get Menu Item Performance & Time-of-Day Sell-Through

Role: MANAGER, ADMIN

GET

GET /api/v1/analytics/item-performance?startDate=2026-07-01&endDate;=2026-07-18&outletId;=&category;=Starters

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "period": "2026-07-01 to 2026-07-18",
  "totalRevenue": 284500.0,
  "totalOrdersProcessed": 1420,
  "topSellingCategory": "Starters",
  "averageCheckAmount": 200.35,
  "dineInVsTakeawayRatio": "70:30"
}
```

5. Get Food Cost & Theoretical vs Actual Variance Report

Role: MANAGER, ADMIN

GET

GET /api/v1/analytics/food-cost?outletId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "period": "2026-07-01 to 2026-07-18",
  "totalRevenue": 284500.0,
  "totalOrdersProcessed": 1420,
  "topSellingCategory": "Starters",
  "averageCheckAmount": 200.35,
  "dineInVsTakeawayRatio": "70:30"
}
```

6. Get Table Turn Time & Occupancy Metrics

Role: MANAGER, ADMIN

GET

GET /api/v1/analytics/table-analytics?outletId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "period": "2026-07-01 to 2026-07-18",
  "totalRevenue": 284500.0,
  "totalOrdersProcessed": 1420,
  "topSellingCategory": "Starters",
  "averageCheckAmount": 200.35,
  "dineInVsTakeawayRatio": "70:30"
}
```

7. Get Swiggy / Zomato / Direct Channel Aggregator Breakdown

Role: MANAGER, ADMIN

GET

GET /api/v1/analytics/aggregator-report?outletId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "totalAggregatorOrders": 124,
  "swiggyRevenue": 34500.0,
  "zomatoRevenue": 28900.0,
  "cancellationRate": 1.2
}
```

8. Get Customer Retention & Feedback Score Dashboard

Role: MANAGER, ADMIN

GET

GET /api/v1/analytics/customer-analytics

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
{
  "customerId": "cust-881",
  "totalVisits": 14,
  "totalLifetimeSpend": 12400.0,
  "averageOrderValue": 885.7,
  "loyaltyPointsBalance": 240,
  "isVip": true
}
```

9. Get Staff Productivity & Shift Revenue Metrics

Role: MANAGER, ADMIN

GET

GET /api/v1/analytics/staff-productivity?outletId=

Headers: Authorization: Bearer

Response Output: **Status 200 OK**

```
[
  {
    "id": "stf-101",
    "name": "Captain Rahul",
    "role": "CAPTAIN",
    "section": "Main Hall",
    "employmentType": "FULL_TIME",
    "isActive": true
  }
]
```